INCH-POUND

MIL-DTL-32628/3
11 April 2019

DETAIL SPECIFICATION SHEET

BACKSHELL, ELECTRICAL CONNECTOR, WIDE SPRING BAND, HEAVY DUTY

This specification is approved for all Departments and
Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32628.

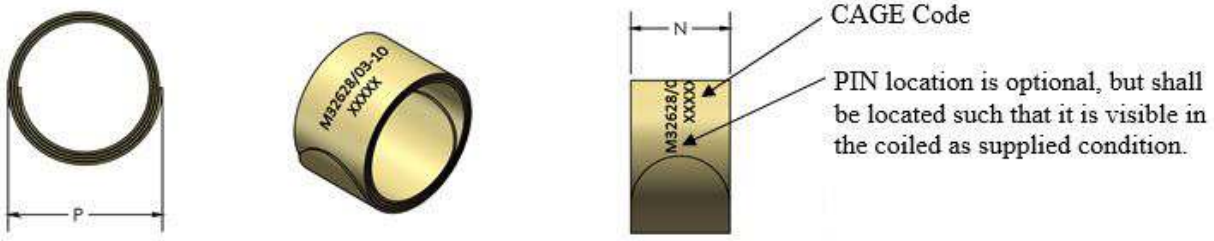


FIGURE 1. Spring configuration (M32628/03-10).

TABLE I. Spring band part number, material, and size.

M32628 /03-XX Part number	Material	Size	
		P Diameter 1/ (±.030 inch)	N Width 1/ (±.030 inch)
-10	Stainless steel in accordance with ASTM A666	0.36	0.37
-11		0.75	
-12		1.05	
-13		1.50	
-20	Non-magnetic cobalt alloy in accordance with SAE AMS5876	0.36	
-21		0.75	
-22		1.05	
-23		1.50	

1/ See figure 1.

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REQUIREMENTS:

1. Dimensions are in inches.
2. The spring band outline configuration is optional within the dimensions shown (see figure 1).
3. Weight. See table II.

TABLE II. Weight.

M32628/03-XX Part number	Weight (1000 count) (lbs.)
-10, -20	2.57
-11, -21	10.60
-12, -22	20.08
-13, -23	38.16

4. Braid retention pull force. See table III.

TABLE III. Braid retention pull force.

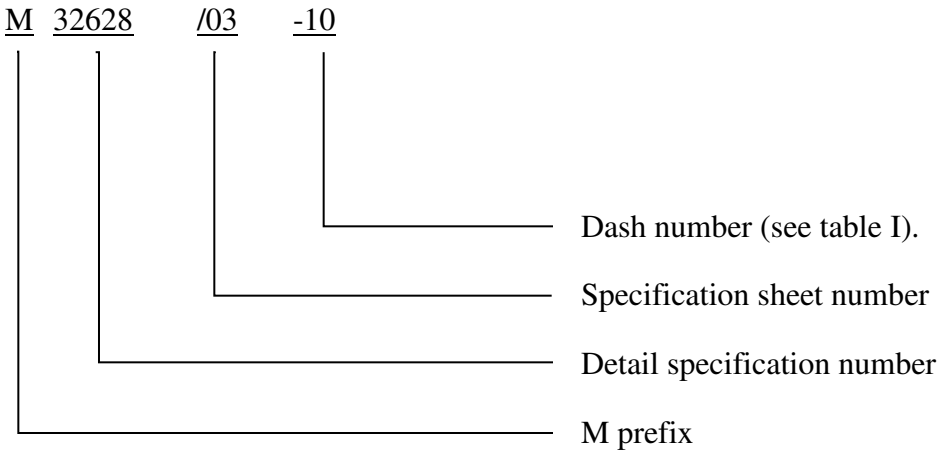
M32628/03 – XX Part number	Cable Entry	A DIA (Maximum) 1/ (inches)	Braid retention pull force (lbs.)
-10, -20	02	.375	100
-10, -20	03	.500	
-11, -21	04	.625	
-11, -21	05	.750	250
-11, -21	06	.875	
-12, -22	07	1.000	
-12, -22	08	1.125	
-12, -22	09	1.250	300
-13, -23	10	1.375	
-13, -23	11	1.500	
-13, -23	12	1.625	
-10, -20	32	.438	100
-10, -20	33	.562	
-11, -21	34	.688	
-11, -21	35	.812	250
-11, -21	36	.938	
-12, -22	37	1.062	
-12, -22	38	1.188	

1/ See figure 3 and table VI.

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5. Part or Identification Number (PIN):

Example:



EXAMINATIONS AND TESTS:

1. First article inspection and conformance inspection testing shall consist of the examinations and tests of MIL-DTL-32628 with the following modifications.

2. First article test samples (see tables IV and V). ISOAS150NF0802-5S, ISOAS150NF2007-5S and ISOAS150NF2810-5S backshells shall be used for first article testing of M32628/03-10, -12, -13 or M32628/03-20, -22 and -23 spring bands, respectively.

- For backshells ISOAS150NF0802-5S, use 34-0500 diameter shield braid (part number AA59569R34T0500),

- For backshells ISOAS150NF2007-5S, use 30-1000 diameter shield braid (part number AA59569R30T1000).

- For backshells ISOAS150NF2810-5S,use 30-1375 diameter shield braid (part number AA59569R30T1375).

- A total of three sample assemblies (braid retention, thermal aging and re-installation testing) plus one spare backshell, shield braid and spring band of each size and material for M32628/03-10, -12, -13 (stainless steel) or M32628/03-20, -22 and -23 (Non-magnetic cobalt) spring bands, depending on spring band material being procured. See table I.

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TABLE IV. First article test samples.

Sample	Backshell	Shield braid	Spring band
Braid retention			
1	ISOAS150NF0802-5S	AA59569R34T0500	M32628/03-10
2	ISOAS150NF2007-5S	AA59569R30T1000	M32628/03-12
3	ISOAS150NF2810-5S	AA59569R30T1375	M32628/03-13
Thermal aging			
4	ISOAS150NF0802-5S	AA59569R34T0500	M32628/03-10
5	ISOAS150NF2007-5S	AA59569R30T1000	M32628/03-12
6	ISOAS150NF2810-5S	AA59569R30T1375	M32628/03-13
Re-installation test			
7	ISOAS150NF0802-5S	AA59569R34T0500	M32628/03-10
8	ISOAS150NF2007-5S	AA59569R30T1000	M32628/03-12
9	ISOAS150NF2810-5S	AA59569R30T1375	M32628/03-13

TABLE V. Test sample components (including one spare).

Quantity	Component
4	ISOAS150NF0802-5S Backshell
4	ISOAS150NF2007-5S Backshell
4	ISOAS150NF2810-5S Backshell
4	AA59569R34T0500 Shield braid (6 inches each)
4	AA59569R30T1000 Shield braid (6 inches each)
4	AA59569R30T1375 Shield braid (6 inches each)
4	M32628/03-10 or M32628/03-20 Spring band
4	M32628/03-12 or M32628/03-22 Spring band
4	M32628/03-13 or M32628/03-23 Spring band

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3. Braid retention. Shield braid shall not pull out and spring band slippage shall not exceed .025 inch when measured from a fixed point. Braid breakage during testing shall not be considered as a failure. One complete assembly for each backshell size with applicable shield braids and spring bands (see table I) shall be tested for a total of three complete assemblies.

a. ISOAS150NF0802-5S backshell assembly assembled with shield braid part number AA59569R34T0500, and spring band M32628/03-10 or -20, pull the braid at a rate of one inch per minute to a force of 100 pounds (see table III).

b. ISOAS150NF2007-5S backshell assembly assembled with shield braid part number AA59569R30T1000, and spring band M32628/03-12 or -22, pull the braid at a rate of one inch per minute to a force of 250 pounds (see table III).

c. ISOAS150NF2810-5S backshell assembly assembled with shield braid part number AA59569R30T1375, and spring band M32628/03-13 or -23, pull the braid at a rate of one inch per minute to a force of 300 pounds (see table III). The shield braid shall not pull out.

NOTES:

1. Intended use. M32628/03 spring bands are intended to be used with Isodyne Inc. connector backshells (see table VI) and braid shield splice. The heavy-duty spring band has increased braid retention pull force for higher shock and vibration operating environments.

2. Backshell compatibility. Spring band part number compatible with the Isodyne Inc. backshells and shield splice. Spring band part number compatible backshells are shown in table VI. Installation of the spring band on the backshells shown in table VI have been validated. Using the spring band on other backshells or accessories may not provide the specified performance requirements of MIL-DTL-32628 spring bands.

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TABLE VI. Commercial backshell compatibility.

Spring band M32628/03 -XX Material/size 1/	Backshell basic part number, shield termination area diameter and cable entry 2/							
	101		150, 155, 160, 165, 170, 180, 750, ST		200		205	
	Cable entry	A DIA REF 3/ (inches)	Cable entry	A DIA REF 3/ (inches)	Cable entry	A DIA REF 3/ (inches)	Cable entry	A DIA REF 3/ (inches)
-10, -20	02	.375	02	.375	02	.375	02	.375
-10, -20	03	.500	03	.500	03	.500	03	.500
-11, -21	04	.625	04	.625	04	.625	04	.625
-11, -21	05	.750	05	0.750	05	.750	05	.750
-11, -21	06	.875	06	0.875	06	.875	---	---
-12, -22	07	1.000	07	1.000	07	1.000	---	---
-12, -22	08	1.125	08	1.125	08	1.125	---	---
-12, -22	09	1.250	09	1.250	---	---	---	---
-13, -23	10	1.375	10	1.375	---	---	---	---
-13, -23	11	1.500	11	1.500	---	---	---	---
-13, -23	---	---	12	1.625	---	---	---	---
-10, -20	32	.438	32	0.438	32	.438	32	.438
-10, -20	33	.562	33	0.562	33	.562	33	.562
-11, -21	34	.688	34	0.688	34	.688	---	---
-11, -21	35	.812	35	0.812	35	.812	---	---
-11, -21	36	.938	36	0.938	36	.938	---	---
-12, -22	37	1.062	37	1.062	---	---	---	---
-12, -22	38	1.188	38	1.188	---	---	---	---

1/ -1X = Stainless steel spring band; -2X = Non-magnetic cobalt alloy spring band.
2/ See figure 2 – cable entry designator.
3/ See figure 3. Note that in addition to the A dimension DIA the thickness of the braid shield must be considered. Shield braid thickness was accounted for in determining appropriate spring band size.

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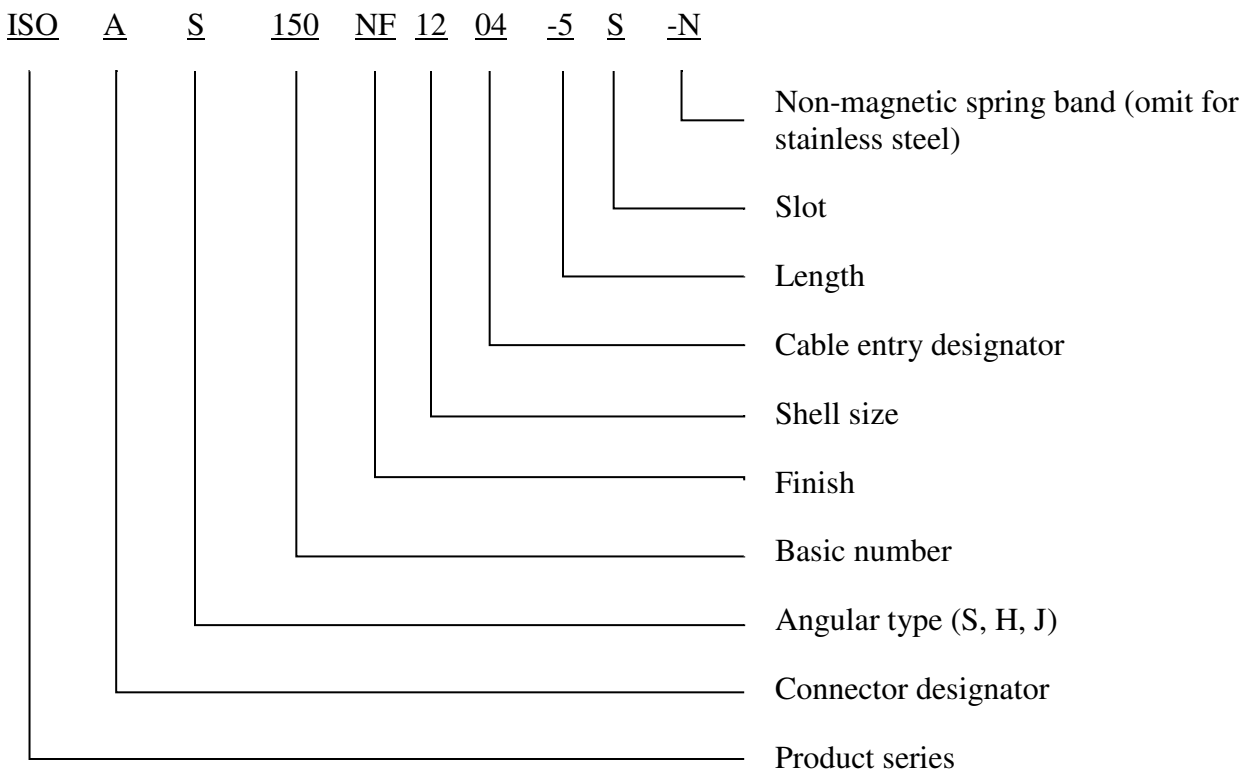


FIGURE 2. Part number example (Isodyne).

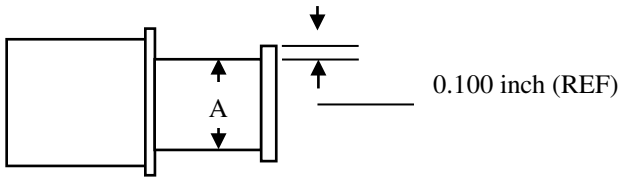


FIGURE 3. Typical commercial backshell accessory.

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3. Isodyne backshells can be obtained from: Isodyne Inc, 7706 E. Osie, Wichita, KS 67207.

4. Spring installation guidance. See MIL-DTL-32628.

5. Spring band application range. Table VII provides the spring band recommended minimum/maximum "A" dimension, outer diameter of the shield termination area for which the spring band may be used. Using that information, the compatible backshell can be selected from table VI.

TABLE VII. Spring band diameter range.

M32628/03 – XX 1/ Part number	A OD Range 2/ (Inches)	
	Min	Max
-10	0.375	.562
-11	.625	.875
-12	.938	1.250
-13	1.375	1.625
-20	.375	.562
-21	.625	.875
-22	.938	1.250
-23	1.375	1.625

1/ -1X = Stainless steel spring band; -2X = Non-magnetic cobalt alloy spring band. See figure 4.
2/ OD range is for information only. Design and finish of the backshell shield termination configuration (finish, knurling, banding shoulder geometry and dimension, etc.) are also important for proper shield termination. Note that in addition to the OD dimension the thickness of the braid shield must be considered. Shield braid thickness was accounted for in determining appropriate spring band size.

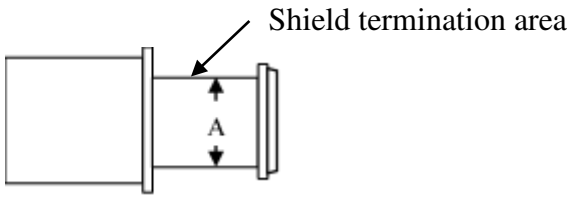


FIGURE 4. Shield termination/banding platform.

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Referenced documents. In addition to MIL-DTL-32628, this specification sheet references:

ASTM A666 SAE AMS5876

Custodians:

Army - CR
Navy - AS
Air Force - 85

Preparing activity:

Navy-AS

Project 5935-2019-017

Review activities:

Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS

Industry association:

SAE AE-8C1

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